A.T.L.A.S. (AUTOMATED TIMETABLING AND LOCATIONS ALLOCATION SYSTEM): A WEB-BASED ACADEMIC SCHEDULING APPLICATION

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Chapter 1

INTRODUCTION

Background of the Study

Educational institutions globally are digitizing core functions like timetabling to improve efficiency, as traditional manual scheduling is highly prone to human errors, double-bookings, and misallocated resources (Nsulangi, P. T., et. al., 2024). This has driven a shift toward automated optimization, seen in institutions like Indonesia's SMPN 1 Jombang (Prakasa, T. A. D., et. al., 2024). Because automated systems reduce errors and easily adapt to dynamic changes (Nsulangi, P. T., et. al., 2024), they are increasingly viewed as a promising, long-term solution to administrative bottlenecks worldwide (Shomoye, M. A. & Mosaku, A. O., 2025).

While international institutions utilize fully automated systems, many Philippine schools still rely on computerizing manual processes via Microsoft Excel and Google Sheets, as seen in the timetabling models of the University of Baguio (Tayong, A. A., & Roma, G. M., 2023) and the City College of Angeles (Yee, B. E. D., et. al., 2024). Although studies confirm that systematized timetabling optimizes resource allocation (Romaguera, D., et. al., 2024), spreadsheet tools remain anchored in manual processes. They successfully computerize data but fail to independently optimize scheduling, still requiring tedious manual input to generate new timetables.

This administrative bottleneck is clearly evident in local public Junior High Schools (JHS), specifically Enriqueta Montilla de Esteban Memorial High School - Main and Hinigaran National High School. Here, class allocation requires balancing complex

physical parameters (like room capacities) and crucial soft constraints often ignored by manual methods, such as minimizing faculty travel time and preventing teacher burnout. Students are also impacted; without real-time communication, last-minute room changes go uncommunicated, causing missed classes. Furthermore, the deployment of automated systems in local public schools is frequently hindered by unstable internet infrastructure, rendering purely cloud-dependent solutions unreliable during connectivity drops. The critical research gap lies in the rigidity of existing automated systems, which are typically hardcoded for a single school. Therefore, there is a clear need for a configurable, centralized algorithm tailored for the JHS level that adapts to diverse priorities while addressing physical logistics and human well-being.

To address this gap, this study proposes A.T.L.A.S. (Automated Timetabling and Locations Allocation System), a generic, mobile-responsive web application designed to balance competing scheduling goals for Junior High Schools. Deploying the system simultaneously at Enriqueta Montilla de Esteban Memorial High School and Hinigaran National High School as pilot sites will validate its configurable, non-hardcoded architecture. A.T.L.A.S. features a priority management module for administrators to dynamically configure rules, such as balancing specialized room matching against faculty workload limits. While the core engine automatically generates conflict-free timetables, a visual drag-and-drop interface allows for seamless manual review and fine-tuning. Finalized schedules are then deployed to a real-time, synchronized dashboard for faculty and students, bridging the communication gap. Finally, the software will undergo rigorous ISO 25010 evaluation to ensure reliability, performance, and stakeholder trust.

Objective of the Study

The general objective of the study aims to design and develop a mobile-responsive web application titled "A.T.L.A.S. (Automated Timetabling and Locations Allocation System): A Web-Based Academic Scheduling Application".

Specifically, this study aims to:

1. design and develop a web and mobile application with the following technical features:
    1.1. web-based administrative portal with drag-and-drop scheduling interface.
    1.2. configurable scheduling priority management module.
    1.3. mobile faculty authentication with schedule preference submission.
    1.4. automated timetable generation system.
    1.5. mobile-responsive student portal with real-time synchronized schedule dashboard.
    1.6. offline-first web application with online synchronization and real-time update capability
2. test the functionality of the features mentioned above
3. evaluate the system using ISO 25010 standardize evaluation tool in terms of:
    3.1. Functionality Suitability
    3.2. Performance Efficiency
    3.3. Usability
    3.4. Compatibility
    3.5. Reliability
    3.6. Security

3.7. Maintainability

3.8. Portability

Significance of the Study

The A.T.L.A.S. (Automated Timetabling and Locations Allocation System): A Web-Based Academic Scheduling Application is designed to be significant and beneficial to the following groups:

The Academic Schedulers (Persons-in-Charge). This study provides a practical, automated solution that directly resolves the administrative bottlenecks of manual timetabling. By replacing spreadsheet data entry with an algorithmic engine and a visual drag-and-drop interface, the research demonstrates how automated priority management can instantly generate conflict-free schedules, saving schedulers weeks of productivity.

The School Faculty and Teachers. The implementation of this system proves the critical importance of integrating human-centric "soft constraints" into educational technology. Instead of randomized schedules, the system actively minimizes physical travel time between distant classrooms and prevents consecutive, heavy teaching loads, providing a working model of how digital tools can protect faculty well-being and prevent burnout.

The Junior High School Students. Students benefit directly from the system's real-time, synchronized, mobile-friendly dashboard. The study demonstrates how eliminating uncommunicated, last-minute room changes fosters a more organized

academic environment, ensuring that students are consistently and accurately allocated to appropriately sized, specialized learning facilities.

The Educational Institutions (Schools). For the participating pilot institutions, Enriqueta Montilla de Esteban Memorial High School - Main and Hinigaran National High School, this research serves as a strategic blueprint for institutional digital transformation. It offers a tested framework showing how implementing "hard constraints" in a centralized algorithm guarantees that physical facilities, specialized laboratories, and room capacities are maximized efficiently without double-bookings or wasted space.

Future IT and Computer Science Researchers. This study serves as a valuable literature reference for future researchers exploring multi-objective optimization algorithms. By detailing a generic, highly configurable architecture and subjecting it to the rigorous ISO 25010 software quality framework, this research provides a foundational academic baseline for expanding automated scheduling solutions into Senior High School or Tertiary educational levels.

Scope and Limitation of the Study

This study covers the design, development, and evaluation of A.T.L.A.S. (Automated Timetabling and Locations Allocation System), a mobile-responsive web application designed to automate academic scheduling for Junior High Schools. Pilot testing will be conducted exclusively at Enriqueta Montilla de Esteban Memorial High School - Main and Hinigaran National High School. The system features a web-based administrative dashboard where schedulers can set specific priorities such as room

capacities, laboratory requirements, and teacher well-being, and use a visual, interactive grid with drag-and-drop capabilities to manually fine-tune room allocations. To ensure the schedule is accurate, teachers are provided with a mobile-friendly portal to directly submit their availability and health preferences before the algorithm runs. Using this collected data, the system automatically generates a conflict-free timetable to prevent double-bookings. Finally, students can access a synchronized, mobile-responsive platform to view real-time updates of their finalized schedules and easily download them as images.

Despite its automated capabilities, the system operates strictly within specific boundaries and is not intended to replace the human administrator. Rather, it serves as a supportive tool to make their work more efficient, reliable, and productive. Recognizing that stable internet access cannot always be guaranteed in school environments, A.T.L.A.S. is designed to function fully offline by default, with internet connectivity serving as an enhancement rather than a requirement. This ensures the application loads from local device storage by default, remaining fully accessible without an active connection. The system requires no installation or download and is accessed entirely through a web browser. Upon a user's first visit, the browser automatically saves a working copy of the system on the device, and from that point forward, all respective user portals load directly from the device first on every visit, regardless of whether an internet connection is present. Any actions completed offline, such as a teacher submitting their preferences or an administrator adjusting a schedule, are saved on the device and automatically sent to the main server once a connection is restored. Schedules

that have been viewed at least once are similarly stored on the device for offline access. However, generating the automated timetable and receiving published schedule updates both require an active internet connection, as these processes depend on the central server where all school data is stored. Functionally, the system's ability to generate a conflict-free schedule still relies entirely on complete and accurate data entry. If the administrator enters conflicting rules or if teacher inputs are missing, the algorithm's performance will be compromised. Furthermore, it strictly functions as a scheduling tool and does not process student enrollment, grades, or attendance, nor does it factor in student preferences, as the student portal is strictly a view-only interface without chat or editing features.

Definition of Terms

For a clearer understanding of the study, the following terms are defined both contextually based on existing literature and operationally as they are used in the context of A.T.L.A.S.

Automation. Contextually, this refers to the use of intelligent algorithms and computerized systems to automatically construct schedules, resolve overlaps, and allocate resources with minimal human intervention (Shomoye & Mosaku, 2025).

Operationally, in this study, it refers to the system's ability to utilize a computational algorithm to independently generate conflict-free academic schedules, replacing the traditional manual data entry in spreadsheet software.

Background Data Synchronization. Contextually, it refers to an automated process in modern web architecture where an application temporarily stores user actions locally while offline and seamlessly uploads them to a central server without requiring manual user intervention once network connectivity is restored (Venzano Rodriguez, 2022).

Operationally, it is the capability of A.T.L.A.S. to safely save teacher preferences or administrative schedule adjustments locally on the device during an internet outage, and automatically upload that data to the main database the moment the connection returns.

Drag-and-Drop Interface. Contextually, this is a graphical user interface (GUI) interaction technique where users select a virtual object by "grabbing" it and moving it to a different location, facilitating intuitive and semi-automated data adjustments within web-based administrative systems (Garcia-Yañez et al., 2024).

Operationally, it is a visual, interactive dash Guia board feature that allows the school administrator to manually click, move, and seamlessly adjust computer-generated room allocations and time slots for fine-tuning purposes.

Hard Constraints. Contextually, in optimization and educational timetabling problems, these are rigid, essential conditions that must be strictly satisfied for a schedule to be considered valid and mathematically feasible; violating them renders the solution unacceptable (Bulley, Petrovic, & Burke, 2023).

Operationally, these are the absolute, non-negotiable physical rules the scheduling algorithm must satisfy to prevent logistical errors, such as strictly adhering to maximum

room capacities, matching subjects to specialized laboratories, and preventing double-bookings of rooms or teachers.

ISO 25010. Contextually, it is the international software engineering standard widely adopted in recent academic research to rigorously evaluate the functional suitability, performance efficiency, and usability of newly developed educational web applications (Canlas et al., 2021).

Operationally, it is the standardized software quality framework used by the researchers to rigorously evaluate the final system's reliability, performance efficiency, and overall usability during the pilot testing phase.

Mobile-Responsive. Contextually, it is a modern web development architecture that dynamically adjusts a system's layout and functionalities depending on the device's screen size, ensuring seamless accessibility across smartphones and desktops without requiring native application installation (Venzano Rodriguez, 2022).

Operationally, it refers to the technical design architecture of the web application that allows it to automatically adapt and function optimally on smartphone screens for teachers and students, eliminating the need to download a native mobile application (APK).

Parameters. Contextually, these are the measurable factors, variables, or input data sets that define the operating conditions of a heuristic algorithm and guide its computational process to find an optimal scheduling solution (Chen et al., 2023).

Operationally, it represents the specific set of rules, physical room limits, and inputted teacher preferences that dictate the boundaries within which the algorithm must operate to generate a valid schedule.

Priority Management Module. Contextually, this represents a multi-objective weighting mechanism where administrators can assign mathematical significance to different scheduling goals, guiding the algorithm to prioritize certain soft constraints over others during generation (Oyelere et al., 2024).

Operationally, it is a configuration feature within the administrative portal that allows the scheduler to dynamically set and rank the importance of different scheduling rules, ensuring the algorithm aligns with the specific needs of the pilot schools.

Real-Time Synchronization. Contextually, it is the automated process of continuously coordinating and updating data across a distributed system simultaneously, ensuring immediate consistency across all viewing interfaces and preventing data silos (Yee et al., 2024).

Operationally, it is the capability of the system to instantly update the faculty and student view-only dashboards the exact moment an administrator finalizes a schedule or makes a manual room adjustment.

Soft Constraints. Contextually, these represent highly desirable preferences in an optimization problem that are not strictly mandatory; violating them does not invalidate the timetable mathematically, but satisfying them significantly improves the overall quality and human acceptance of the schedule (Bulley, Petrovic, & Burke, 2023).

Operationally, these are the highly desired, human-centric conditions the algorithm attempts to satisfy to protect teacher well-being, such as minimizing physical travel time between distant classrooms and preventing heavy, continuous teaching loads without breaks.

Timetabling. Contextually, it is defined as the allocation, subject to specific physical and human constraints, of given resources to objects being placed in space-time, arranged in such a way as to satisfy a set of desirable educational objectives as nearly as possible (Bulley, Petrovic, & Burke, 2023).

Operationally, it is the complex administrative process of plotting and allocating teachers, junior high school students, subjects, and physical classrooms into specific time slots without encountering logistical conflicts.

Chapter 2

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter presents the related literature and studies from both local and foreign sources. Those were included in this chapter helps in familiarizing information that are relevant and similar to the present study.

Visual Interactive Interfaces and Drag-and-Drop Functionality in Scheduling

While algorithmic generation successfully resolves the heavy computational burden of academic scheduling, strict mathematical models often lack the nuanced flexibility required by human administrators. To bridge this gap, modern educational technology emphasizes pairing automatic engines with intuitive, visual interfaces. Research indicates that integrating drag-and-drop grids with color-coded alerts allows administrators to dynamically refine schedules, reducing overall scheduling time by up to 75% compared to conventional methods (Ghazal et al., 2025). This deployment of interactive interfaces is a recognized standard across leading commercial and open-source platforms; systems like SchoolTracs, Lantiv Timetabling Turbo, and the FET Timetable Explorer rely heavily on drag-and-drop functionality to empower users to effortlessly adjust classes, resolve edge-case conflicts, and finalize generated schedules without manual data input (SchoolTracs, 2025; Lantiv, 2026; FET, 2025).

Although these existing platforms validate that visual drag-and-drop functionality is an essential industry standard, they frequently present severe architectural and financial limitations for local institutions. Commercial solutions like SchoolTracs and Lantiv

function as expensive, generalized cloud-based products requiring continuous subscriptions, while utilities like FET require complex desktop installations. A.T.L.A.S. integrates this proven interactive methodology to ensure maximum administrative control but applies it within an offline-capable, locally adapted web architecture. Following the automated generation phase, the system provides a specialized drag-and-drop grid that empowers administrators to manually override allocations and fine-tune room assignments. This ensures the critical visual interface remains accessible, cost-effective, and precisely tailored to the target institution's operational constraints without relying on unstable internet connections or external commercial licenses.

Configurable Priority Management and Human-Centric Soft Constraints

The transition from rigid algorithmic scheduling to adaptable, human-centric priority management is essential for institutional fairness and faculty well-being. Recent literature emphasizes that modern timetabling must extend beyond merely preventing spatial conflicts to strategically incorporate soft constraints, such as minimizing idle periods and balancing teaching loads. Studies evaluating complex high school scheduling models demonstrate that mathematically prioritizing these human parameters results in timetables that are significantly more practical and sustainable for educators (Sun & Wu, 2023). Furthermore, by formally defining pre-determined teaching load limits and faculty skills, administrative tools can generate objective assignments that successfully eliminate administrative bias and reduce workplace grievances (Khorbotly & White, 2024).

Within the Philippine educational context, the effectiveness of these automated systems relies heavily on an administrator's ability to control and customize these exact logistical parameters. A recent local evaluation highlighted that empowering administrators to dynamically configure scheduling criteria—rather than relying on rigid, hardcoded constraints—ensures the software successfully meets the specific, evolving needs of an institution (Keh & Sarmiento, 2025). A.T.L.A.S. directly applies this proven standard through its configurable priority management module. By providing a centralized, dynamic dashboard, the system allows school administrators to independently adjust the mathematical weights of various soft constraints, such as maximum teaching hours and department-level rules. This ensures the algorithmic engine consistently adapts to the school's unique policies while maintaining absolute systemic fairness and respecting teacher well-being.

Secure Faculty Portals and Decentralized Data Collection

Beyond secure authentication, the primary function of a dedicated faculty portal is the accurate, decentralized collection of scheduling parameters. Algorithmic timetabling models rely heavily on precise data inputs to maximize the alignment between course offerings and faculty capabilities. Recognizing the time of qualified instructors as a highly valuable academic resource, research argues that directly integrating faculty preferences into the scheduling engine is critical for optimal assignment (Khorbotly & White, 2024). Furthermore, recent technological deployments in higher education demonstrate that providing professors with web-based portals to autonomously enter their availability—

such as preferred days, hours, and subjects—drastically reduces academic programming errors and logistical bottlenecks. By decentralizing this data-entry process, institutions eliminate administrative conflicts and save significant man-hours previously wasted on manual transcription (Garcia Yañez et al., 2024).

A.T.L.A.S. directly addresses these operational necessities by incorporating a secure, mobile-responsive faculty portal linked seamlessly to its central timetable generator. This architecture empowers faculty members to safely and independently submit their teaching availability, preferred schedules, and soft constraints. Once submitted, the system automatically processes these variables during schedule generation to maximize resource efficiency. By ensuring the automated engine is fed accurate, real-time data directly from the instructors, A.T.L.A.S. completely removes the administrator as the middleman for data entry, fulfilling the objective of accurate, error-free constraint gathering and preventing the logistical bottlenecks of manual scheduling.

Automated Timetable Generation and Algorithmic Optimization

Educational timetabling is universally classified as an NP-hard problem, meaning the sheer volume of logistical constraints makes it mathematically infeasible to resolve manually without exhausting administrative resources and consuming hundreds of man-hours (Tan et al., 2021; Yavuz et al., 2023). As secondary education curricula grow in complexity, modern decision support systems must transition from manual plotting to advanced algorithmic optimization to manage escalating workloads. Recent literature demonstrates that utilizing stacked metaheuristic models, such as simulated annealing,

particle swarm optimization (PSO), and two-phase tabu search algorithms, significantly outperforms traditional heuristic methods by rapidly processing complex variables like faculty loading and room capacities (Prakasa et al., 2024; Sun & Wu, 2023; Tassopoulos et al., 2023; Uysal et al., 2025). These computational approaches effectively eliminate redundant calculations and resolve systemic bottlenecks, successfully compressing weeks of manual trial-and-error into a fraction of the execution time.

To navigate the most severe constraints of academic resource allocation, highly scalable automated engines frequently utilize biology-inspired Evolutionary Algorithms, specifically Genetic Algorithms (GA), as their computational backbone. International research and advanced hybrid frameworks prove that by applying computational crossover and mutation techniques, GA excels at iteratively refining complex matrices of room assignments and teacher eligibility, gracefully resolving dense bottlenecks where traditional models fail (Katore et al., 2024; Ranolo et al., 2025; Uma et al., 2024). Furthermore, recent local evaluations validate this architecture, demonstrating that enhanced genetic algorithms successfully optimize spatial resources and eliminate schedule conflicts within the Philippine educational context (Romaguera et al., 2024). A.T.L.A.S. integrates these proven local and international mathematical principles into its core generation engine. By leveraging the adaptive optimization of genetic algorithms, the system systematically processes strict institutional constraints to instantly output a mathematically feasible, conflict-free schedule tailored specifically for the logistical realities of Junior High Schools.

Real-Time Synchronization and Role-Based Student Dashboards

The overall effectiveness of an academic schedule relies not only on its precise mathematical generation but also on its immediate and accurate dissemination to the student body. Usability research on multi-role educational platforms indicates that classifying functionalities through distinct, role-specific dashboards—tailored separately for administrators, teachers, and students—drastically improves system clarity and reduces institutional miscommunication (Li, 2025). Furthermore, centralizing this logistical data into dynamic, mobile-friendly interfaces empowers students to independently track schedules and download timetables, fostering a highly responsive, learner-centered environment that minimizes traditional administrative delays (Dahlan et al., 2024; Pujan, 2025). When these user-centric interfaces are seamlessly connected to a centralized scheduling engine, any logistical modifications or room changes can be instantly distributed across the campus, bridging administrative workflows with end-user accessibility to prevent missed classes and mitigate campus confusion (Uysal et al., 2025).

The necessity of this real-time data distribution is heavily validated by modern commercial systems; platforms like SchoolTracs actively utilize synchronized applications to instantly push schedule updates to dedicated teacher and student dashboards (SchoolTracs, 2025). However, while these commercial Software as a Service (SaaS) products successfully demonstrate the efficacy of role-based syncing, they present severe architectural limitations for local institutions by relying entirely on mandatory cloud connectivity and expensive subscription models. A.T.L.A.S. integrates these established UI/UX best practices and industry-standard synchronization features but applies them

within an environment optimized for local constraints. By deploying distinct, mobile-responsive portals, the system ensures that once a master schedule is generated, faculty and students interact only with the synchronized data relevant to their specific roles. This allows the target institution to achieve high-level scheduling transparency and real-time communication without the financial burden or operational risks of commercial cloud licenses.

Offline-First Architecture and Data Synchronization

Recent developments in educational technology increasingly emphasize the necessity of offline-capable systems, particularly in environments hampered by unreliable internet infrastructure. To guarantee continuous accessibility, modern web applications must reduce their reliance on constant server connectivity by integrating local data caching and browser-based storage mechanisms. Research highlights that this "offline-first" architectural approach allows users to access critical scheduling data quickly and reliably, entirely independent of active server connections (Padshetty et al., 2025; Pujan, 2025). Furthermore, recent offline systems, such as the Campus Genie digital assistant, demonstrate that utilizing localized data storage to bypass continuous cloud connectivity successfully ensures seamless academic operations even in limited-network contexts (Vadivel et al., 2025). This literature confirms that integrating offline data persistence is not just a secondary feature, but a critical requirement for maintaining the operational flow of modern educational platforms.

A.T.L.A.S. directly adopts this resilient architectural standard to mitigate the infrastructure limitations commonly faced by local public schools. By implementing a lightweight, offline-first web application, the system leverages local browser storage to cache timetables and user interfaces upon the initial load. This ensures that school administrators and faculty can reliably view the master schedule and safely queue logistical adjustments even during complete internet outages. To maintain strict data accuracy across the institution without sacrificing this local reliability, the localized foundation is augmented with automatic deferred synchronization. The moment a stable network connection is detected, the application seamlessly synchronizes all queued local modifications with the central server in the background, ensuring that all distributed devices instantly reflect the finalized schedule without ever locking the user out of the system.

Table 1.

Matrix of Related Literatures and Studies

RRL	**Features**					
Studies/System	web-based administrative portal with drag-and-drop scheduling interface.	configurable scheduling priority management module	mobile faculty authentication with schedule preference submission	. automated timetable generation system	mobile-responsive student portal with real-time synchronized schedule dashboard	offline-first web application with online synchronization and real-time update capability

Real-Time Modular Scheduling System for Educational Operations – Ghazal et al. (2025)	✓	✗	✗	✗	✗	✗
Timetabling Turbo 2026 [Computer software] – Lantiv (2026)	✓	✗	✗	✗	✓	✗
FET Timetable Explorer [Computer software] – FET (2025)	✓	✗	✗	✗	✗	✗
Easy drag-and-drop scheduling [Computer software] – SchoolTracs (2025)	✓	✗	✗	✗	✗	✗
A Preference-Based Faculty-Assignment Tool for Course Scheduling Optimization – Khorbotly & White (2024)	✗	✓	✓	✗	✗	✗
AUTOMATED CLASS SCHEDULING SYSTEM FOR AEMILIANUM COLLEGE INC. – Keh & Sarmiento (2025)	✗	✓	✗	✗	✗	✗
Two-phase tabu search algorithm for solving Chinese high school timetabling problems under the new college entrance examination reform – Sun & Wu (2023)	✗	✓	✗	✓	✗	✗
A Preference-Based Faculty-Assignment Tool for Course Scheduling Optimization. 2024 ASEE Annual Conference & Exposition – Khorbotly & White (2024)	✗	✓	✓	✗	✗	✗
Development of a Web-Based Timetabling Software for a Mexican University – Garcia Yañez et al. (2024)	✗	✗	✓	✗	✗	✗
A Course Timetabling Problem for Classroom Usage Minimization – Yavuz et al. (2023)	✗	✗	✗	✓	✗	✗
A Web-based Decision Support System for Managing Course Timetabling in Online Education – Uysal et al. (2025)	✗	✗	✗	✓	✓	✗
A survey of the state-of-the-art of optimisation methodologies in school timetabling problems – Tan et al. (2021)	✗	✗	✗	✓	✗	✗

Two-phase tabu search algorithm for solving Chinese high school timetabling problems under the new college entrance examination reform – Sun & Wu (2023)	✗	✓	✗	✓	✗	✗
An Effective Local Particle Swarm Optimization-Based Algorithm for Solving the School Timetabling Problem – Tassopoulos et al. (2023)	✗	✗	✗	✓	✗	✗
Automating School Timetabling: An Intelligent System Application Using Simulated Annealing – Prakasa et al. (2024)	✗	✗	✗	✓	✗	✗
Dynamic Timetable Generator Using Genetic Algorithm – Katore et al. (2024-2025)	✗	✗	✗	✓	✗	✗
Automated timetable generation for academic institutions – Uma et al. (2024)	✗	✗	✗	✓	✗	✗
Development of a Web-based Course Timetabling System based on an Enhanced Genetic Algorithm – Romaguera et al. (2024)	✗	✗	✗	✓	✗	✗
A Hybrid Approach to Automatic Timetabling Using Self-Organizing Maps, Secure Convex Dominating Sets, and Metaheuristics – Ranolo et al. (2025)	✗	✗	✗	✓	✗	✗
Design and implementation of a cross-platform Student Management System app using OOP principles – Pujan (2025)	✗	✗	✗	✗	✓	✓
A Web-based Decision Support System for Managing Course Timetabling in Online Education – Uysal et al. (2025)	✗	✗	✗	✓	✓	✗
User Experience Design and Evaluation of a Multi-Role School Management Platform – Li (2025)	✗	✗	✗	✗	✓	✗
EffiESGen: A User-centric Web Application for Efficient Exam Scheduling and Collaboration in Education – Dahlan et al. (2024)	✗	✗	✗	✗	✓	✗
Campus Genie: A Secure, Offline NLP-Powered Assistant for College Students – Vadivel et al. (2025)	✗	✗	✗	✗	✗	✓

Design and implementation of a cross-platform Student Management System app using OOP principles – Pujan (2025)	✗	✗	✗	✗	✓	✓
Automated Time-Table Management System – Padshetty et al. (2025)	✗	✗	✗	✗	✗	✓

Synthesis

The review of related literature and systems establishes academic timetabling as a highly complex, NP-hard logistical challenge that consistently exceeds the capacity of manual, heuristic methodologies. To resolve the dense matrix of spatial limitations and human-centric variables, modern educational institutions necessitate advanced computational frameworks; consequently, A.T.L.A.S. employs Evolutionary and Genetic Algorithms as its central generation engine to achieve mathematical optimization in resource allocation. However, current literature emphasizes that algorithmic efficacy is intrinsically dependent on accurate, decentralized data collection. The system fulfills this requirement by integrating secure faculty portals for direct availability input alongside a configurable priority management module, empowering administrators to mathematically weigh institutional soft constraints. Acknowledging that strict computational models cannot invariably anticipate real-world edge cases, A.T.L.A.S. bridges the divide between machine calculation and human oversight through an interactive drag-and-drop grid tailored for manual schedule refinement. Upon finalization, this data is immediately disseminated across the institution via real-time, synchronized, role-based dashboards to guarantee logistical alignment among all stakeholders. Crucially, to adapt these industry-standard capabilities to the infrastructural realities of local Philippine schools, the entire

ecosystem is built upon a resilient, offline-first web architecture. Ultimately, the synthesis of these theoretical models and applied systems validates A.T.L.A.S. as a comprehensive, mathematically sound, and highly accessible solution to the modern academic timetabling dilemma.

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